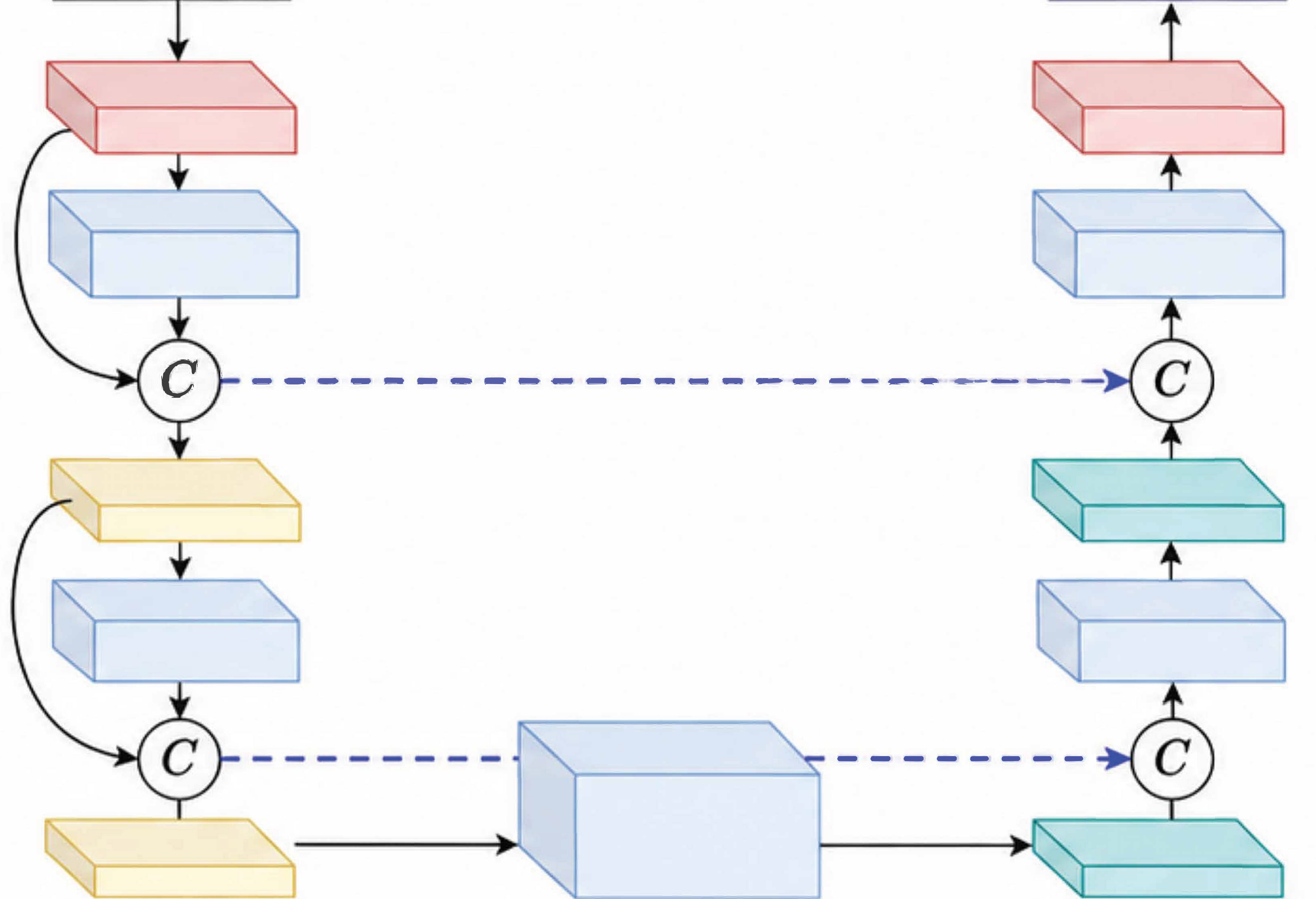
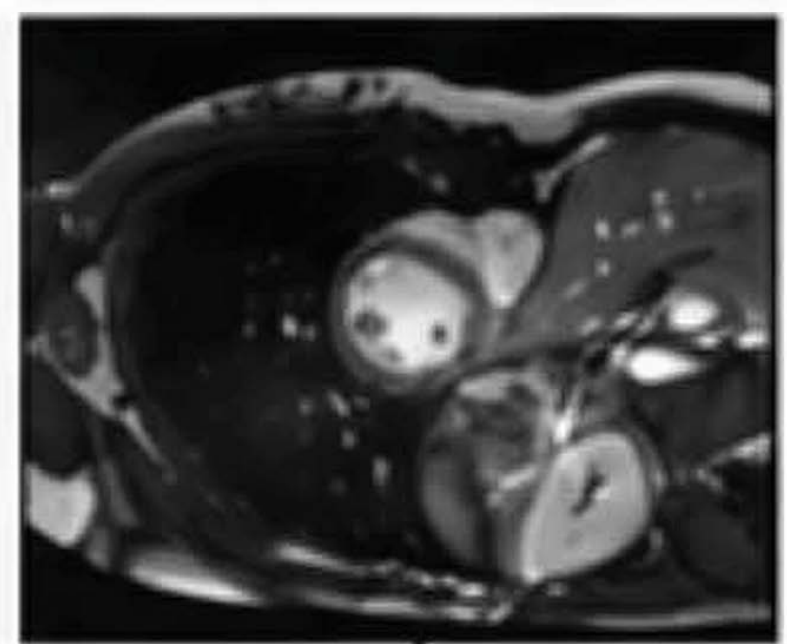


Proposed Lightweight Segmentation Network

Input Image



Convolution Layer



Dense Layer



Concatenation



Transition Down

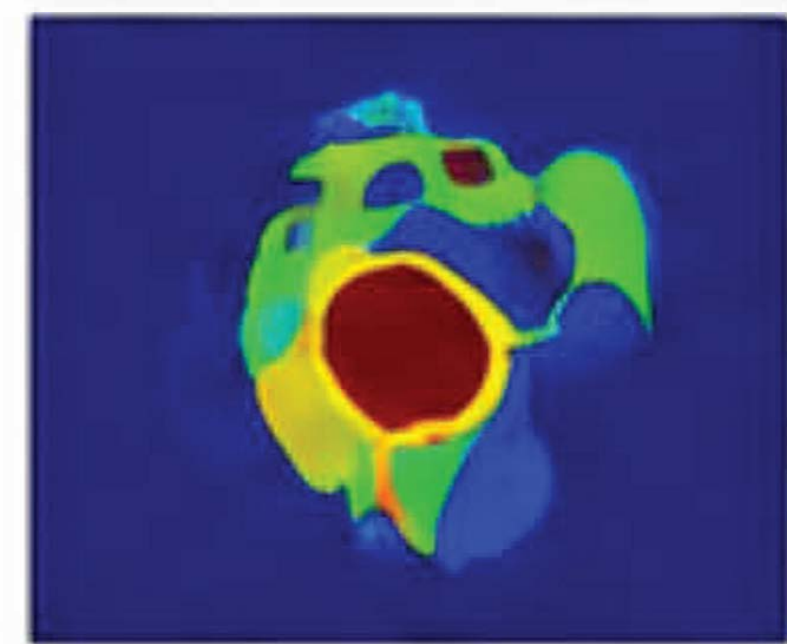


Transition Up



Skip Connection

Segmentation Map



Efficient Deployment on Low-Cost Hardware



Computer

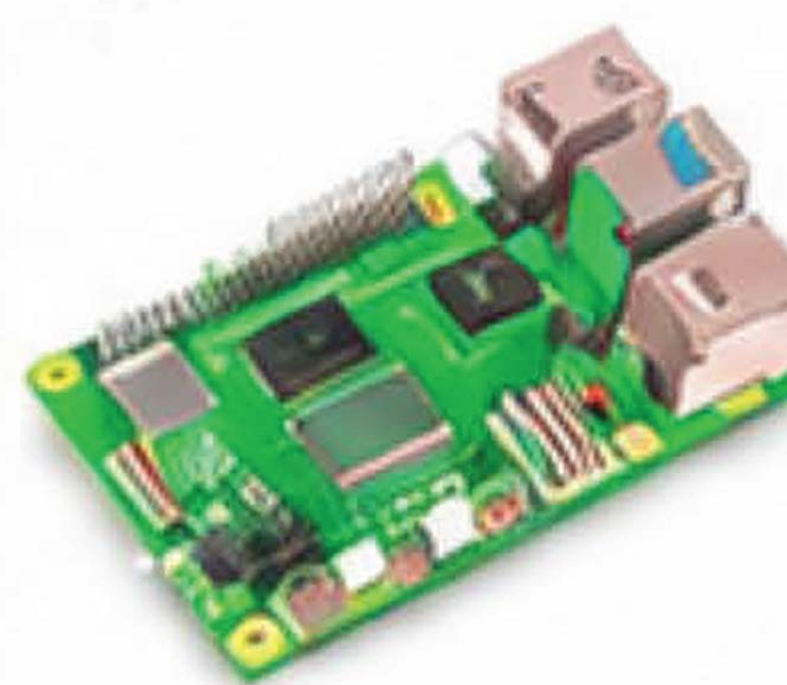
- GPU Inference
- Multi-threading
- Optimized Runtime

Edge Devices



Jetson Nano

- FP16 Inference
- TensorRT Optimization
- Low Power

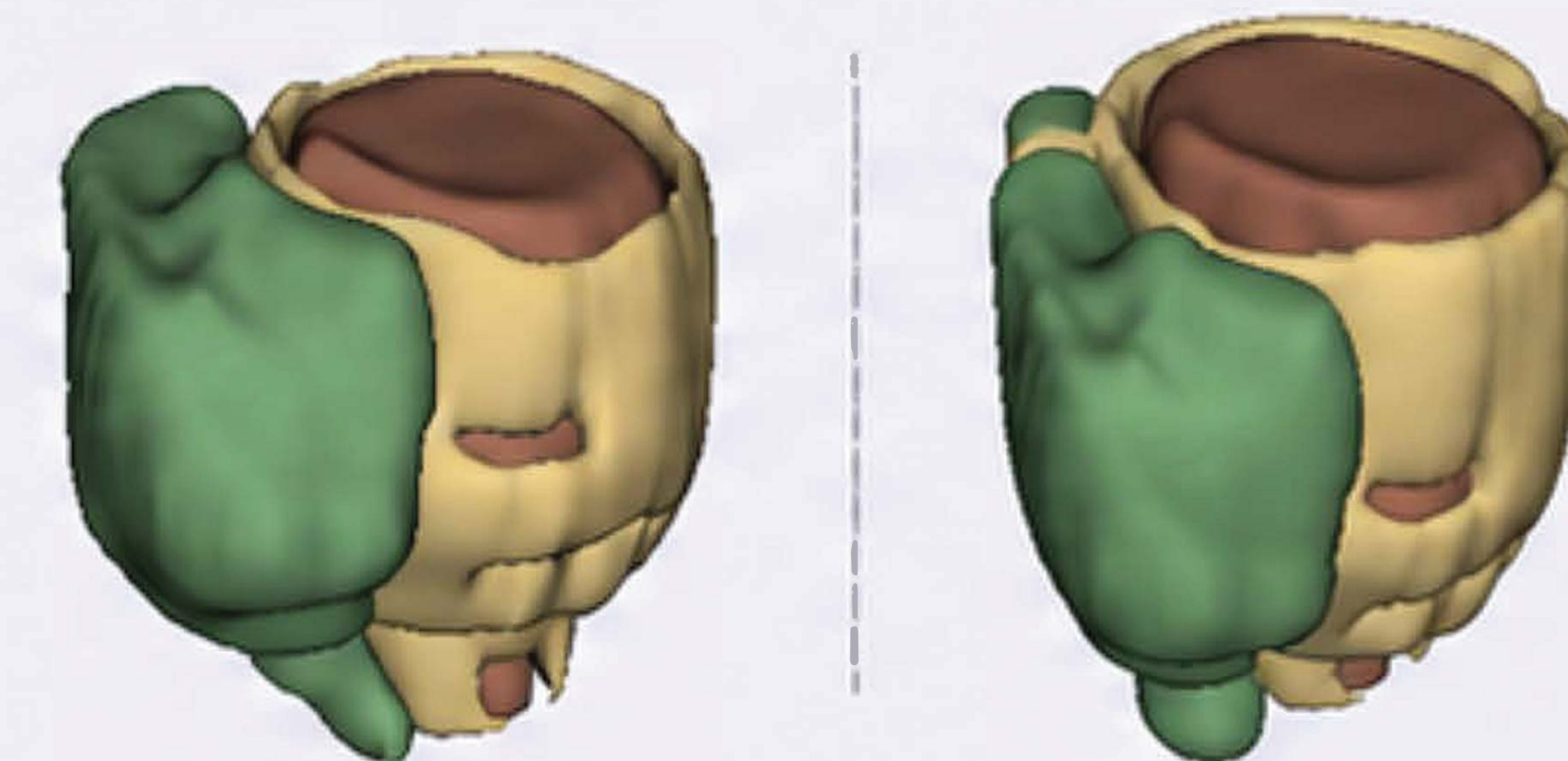


Raspberry Pi

- INT8 Quantization
- Lightweight Inference
- Portable & Low Cost

Outputs and Clinical Integration

3D Volumes



Quantitative Measurements



LV EDV



LV ESV

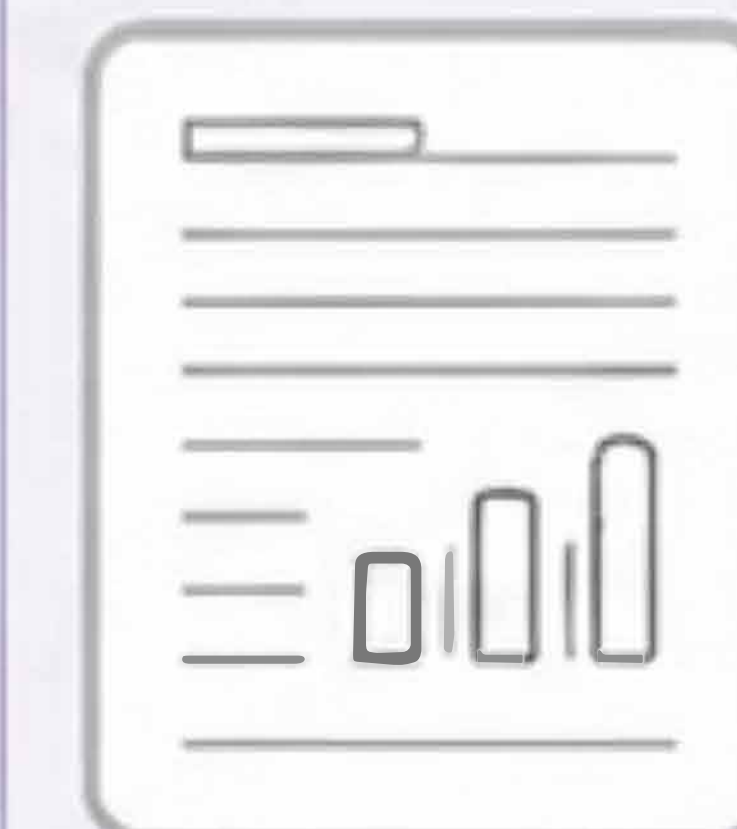


LV EF (%)



Myocardial Mass

Automated Patient Report



- ✓ Segmentation Overview
- ✓ Quantitative Results
- ✓ Visualizations
- ✓ Summary & Findings